Toxicity Predictions (LD_{50}) & Model Interpretations for Selected Drug Combinations

This document presents a set of drug–drug combination test cases generated as part of the project "Deep Graph-based QSAR Modeling with XAI for Interpretable Prediction of Drug–Drug Interaction Toxicity (LD_{50})". For each pair, the model predicts a toxicity value and provides an explanation by highlighting important atoms and structural features in the molecules.

LD_{50} (median lethal dose) refers to the amount of a substance required to cause death in 50% of a test population, and it is commonly used as a measure of toxicity.

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Test Case 01 (Tramadol + Diazepam)

```
Resolved Drug A (name): Tramadol
SMILES A: COc1cccc(C2(O)CCCCC2CN(C)C)c1
Resolved Drug B (name): Diazepam
SMILES B: CN1C(=O)CN=C(c2ccccc2)c2cc(Cl)ccc21

--- Chemical Interpretation ---

Drug A: Tramadol
Drug B: Diazepam

Predicted pLD50: 2.2800
Predicted toxicity level: moderate toxicity

1. Model focus

Drug A important atoms:
- N sp3 degree 3 (atom 15); C sp3 degree 1 3 H (atom 0); aromatic C sp2 de
gree 3 (atom 2)

Drug B important atoms:
```

- aromatic C sp2 degree 2 1 H (atom 10); aromatic C sp2 degree 2 1 H (atom 9); aromatic C sp2 degree 2 1 H (atom 11)

2. Chemical meaning of important atoms

Drug A:
- Atom 15 is a nitrogen atom, which may indicate an amine-like or heteroatom-rich reactive region. Atom 0 is a C atom that may contribute to the local chemical environment. Atom 2 is an aromatic carbon, which may belong to a hydrophobic ring system.

Drug B:
- Atom 10 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 9 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 11 is an aromatic carbon, which may belong to a hydrophobic ring system.

3. Key structural fragments

Drug A:
- C -> a general structural fragment; c -> an aromatic hydrophobic ring; CNC -> an amine-like or nitrogen-containing region

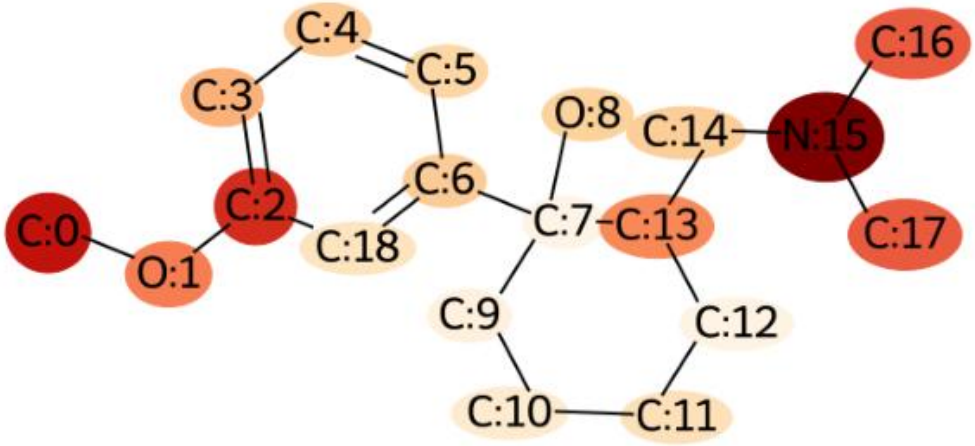
Drug B:
- N -> a nitrogen-containing heterocycle; ccc -> an aromatic hydrophobic ring

4. Interaction-level interpretation

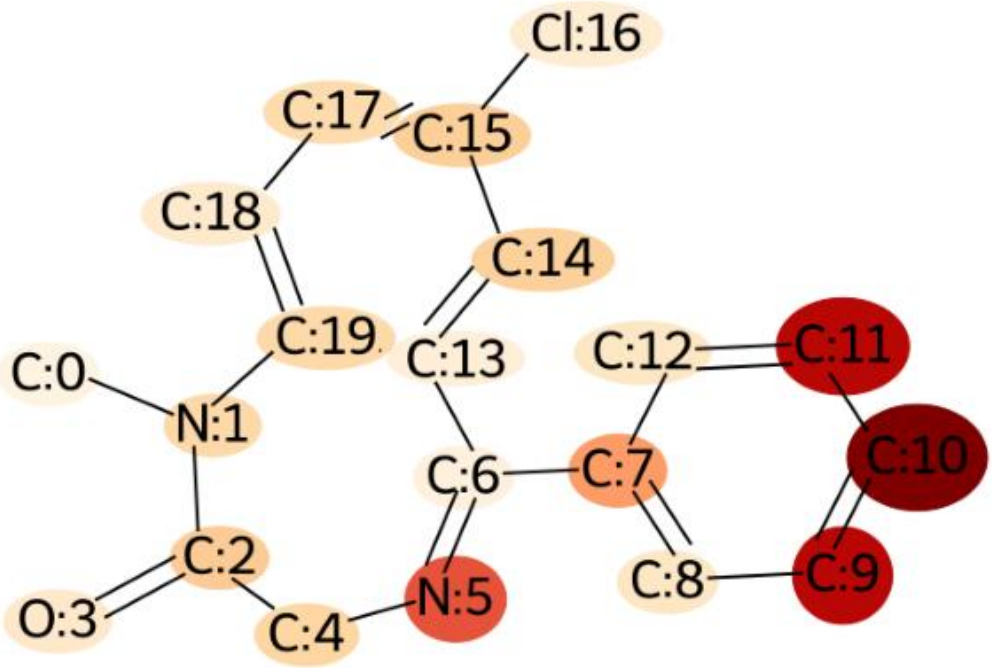
The model highlights a general structural fragment in Tramadol and a nitrogen-containing heterocycle in Diazepam, suggesting that their interaction may influence toxicity through combined structural and chemical effects.

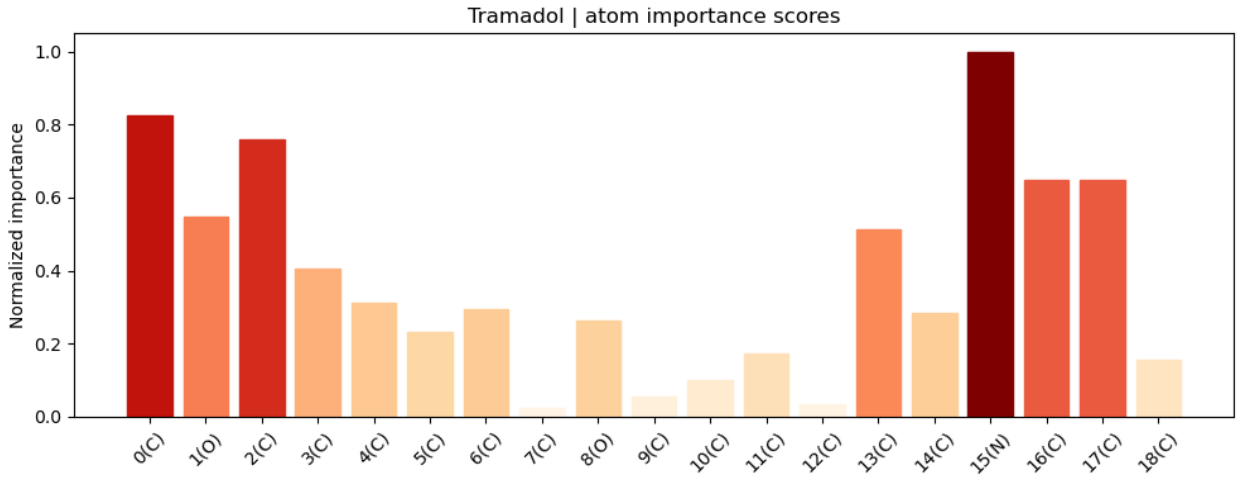
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

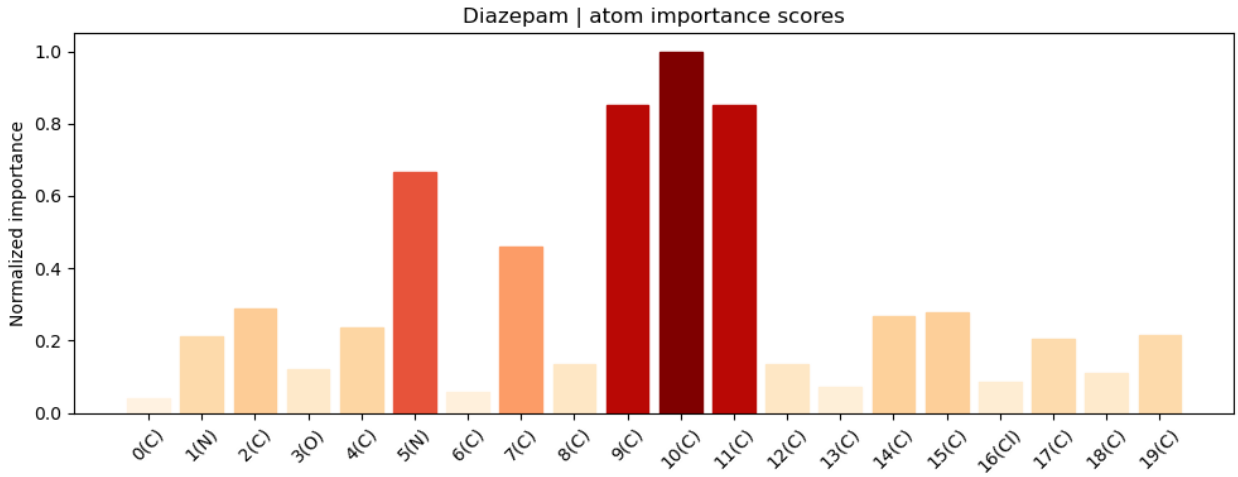
Tramadol | highlighted atom importance



Diazepam | highlighted atom importance







Test Case 02 (Paracetamol + Phenytoin)

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Resolved Drug A (smiles): N-(4-hydroxyphenyl)acetamide
SMILES A: CC(=O)Nc1ccc(O)cc1
Resolved Drug B (smiles): 5,5-diphenylimidazolidine-2,4-dione
SMILES B: O=C1NC(=O)C(c2ccccc2)(c2ccccc2)N1

--- Chemical Interpretation ---

Drug A: N-(4-hydroxyphenyl)acetamide
Drug B: 5,5-diphenylimidazolidine-2,4-dione

Predicted pLD50: 2.6200
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- aromatic C sp2 degree 3 (atom 7); O sp2 degree 1 1 H (atom 8); N sp2 degree 2 1 H (atom 3)

Drug B important atoms:
- ring N sp2 degree 2 1 H (atom 18); ring C sp3 degree 4 (atom 5); ring N sp2 degree 2 1 H (atom 2)

2. Chemical meaning of important atoms

Drug A:
- Atom 7 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 8 is an oxygen atom, which may indicate a polar functional group. Atom 3 is a nitrogen atom, which may indicate an amine-like or heteroatom-rich reactive region.

Drug B:
- Atom 18 is a ring nitrogen, which may influence polarity, basicity, or binding behavior. Atom 5 is a ring carbon, which may contribute to structural rigidity. Atom 2 is a ring nitrogen, which may influence polarity, basicity, or binding behavior.

3. Key structural fragments

Drug A:
- cO -> an aromatic hydrophobic ring

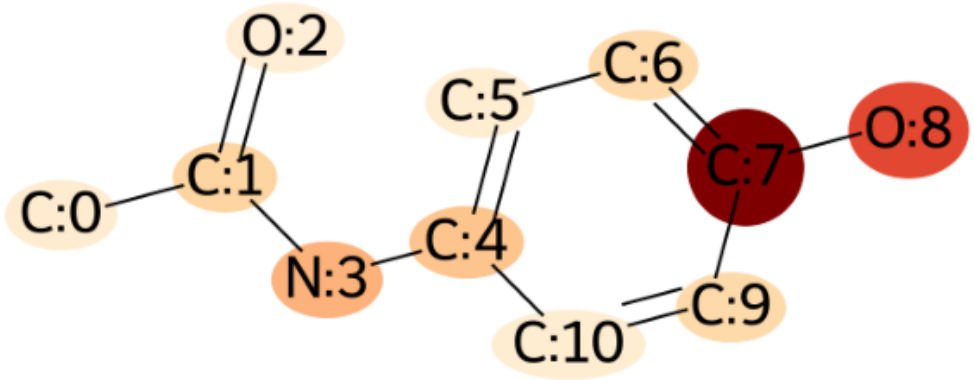
Drug B:
- CN -> a nitrogen-containing heterocycle

4. Interaction-level interpretation

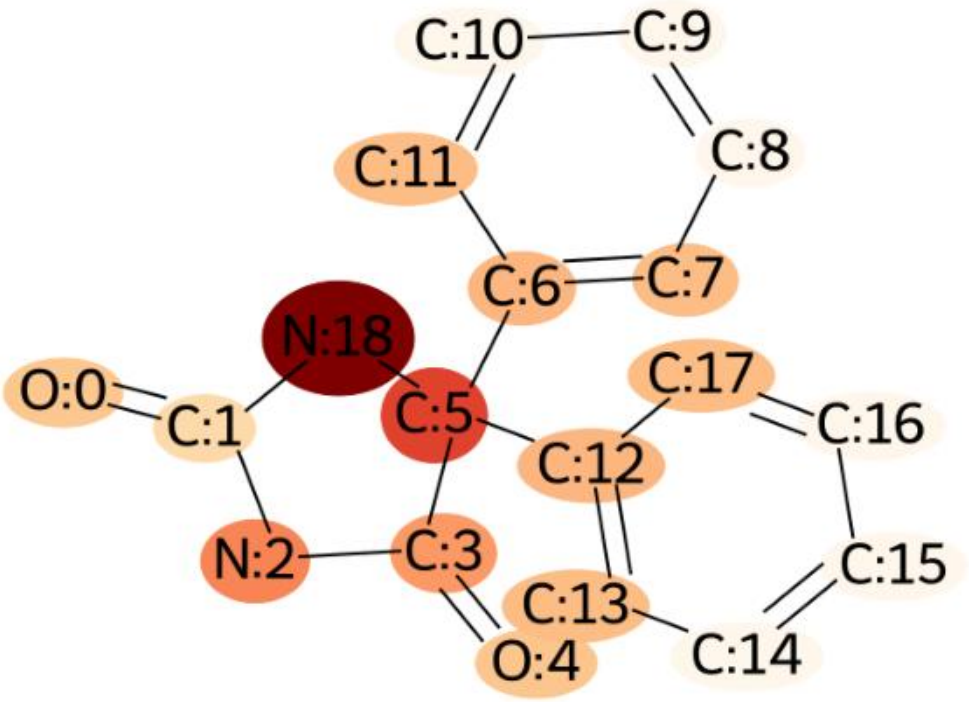
The model highlights an aromatic region in one molecule and a polar or heteroatom-rich region in the other, suggesting that interactions between hydrophobic and polar functionalities may contribute to the observed toxicity.

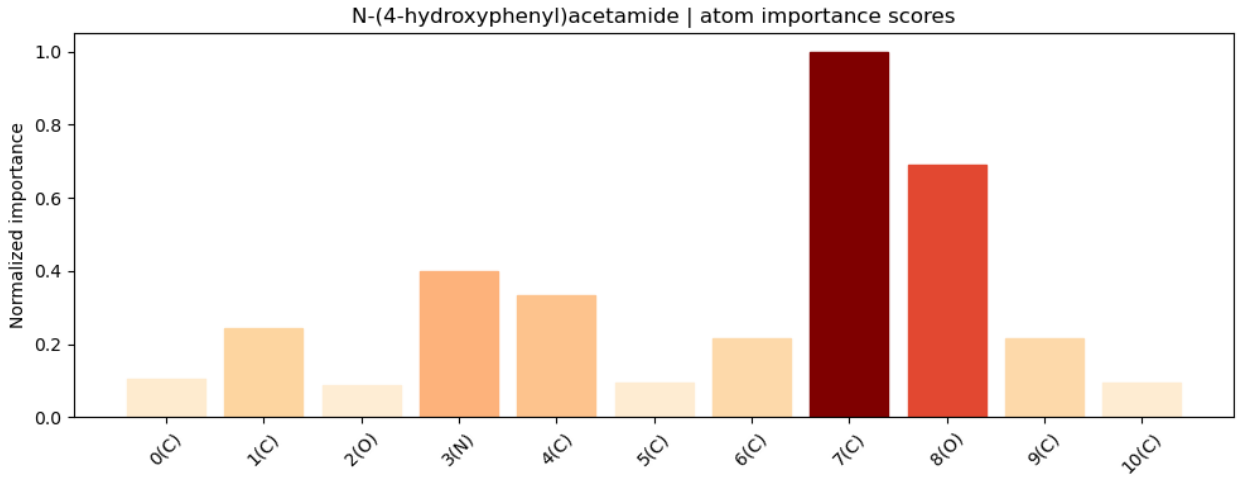
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

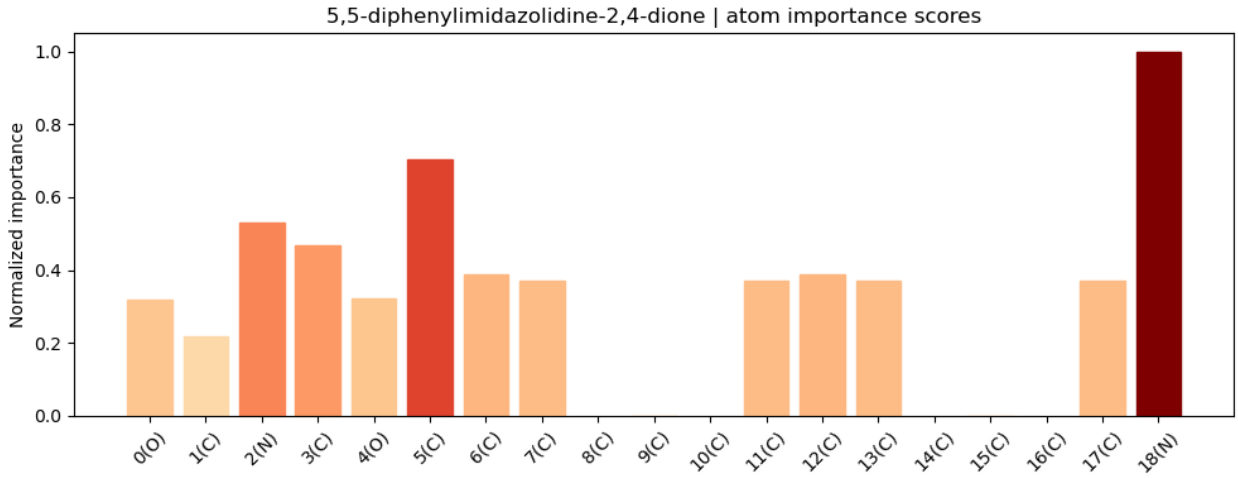
N-(4-hydroxyphenyl)acetamide | highlighted atom importance



5,5-diphenylimidazolidine-2,4-dione | highlighted atom importance







Test Case 03 (Warfarin + Ibuprofen)

```
Enter Drug A common name or SMILES: Warfarin
Enter Drug B common name or SMILES: Ibuprofen
Resolved Drug A (name): Warfarin
SMILES A: CC(=O)CC(c1ccccc1)c1c(O)c2ccccc2oc1=O
Resolved Drug B (name): Ibuprofen
SMILES B: CC(C)Cc1ccc(C(C)C(=O)O)cc1

--- Chemical Interpretation ---

Drug A: Warfarin
Drug B: Ibuprofen

Predicted pLD50: 2.5086
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- C sp3 degree 3 1 H (atom 4); O sp2 degree 1 (atom 22); aromatic C sp2 degree 2 1 H (atom 8)

Drug B important atoms:
- aromatic C sp2 degree 3 (atom 7); C sp3 degree 2 2 H (atom 3); C sp3 degree 3 1 H (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 4 is a highly connected C atom, which may lie in a structurally important region. Atom 22 is an oxygen atom, which may indicate a polar functional group. Atom 8 is an aromatic carbon, which may belong to a hydrophobic ring system.

Drug B:
- Atom 7 is an aromatic carbon, which may belong to a hydrophobic ring system. Atom 3 is a C atom that may contribute to the local chemical environment. Atom 1 is a highly connected C atom, which may lie in a structurally important region.

3. Key structural fragments

Drug A:
- C=O -> an oxygen-rich polar chain or functional group; C -> a general structural fragment; ccc -> an aromatic hydrophobic ring; O -> an oxygen-rich polar chain or functional group

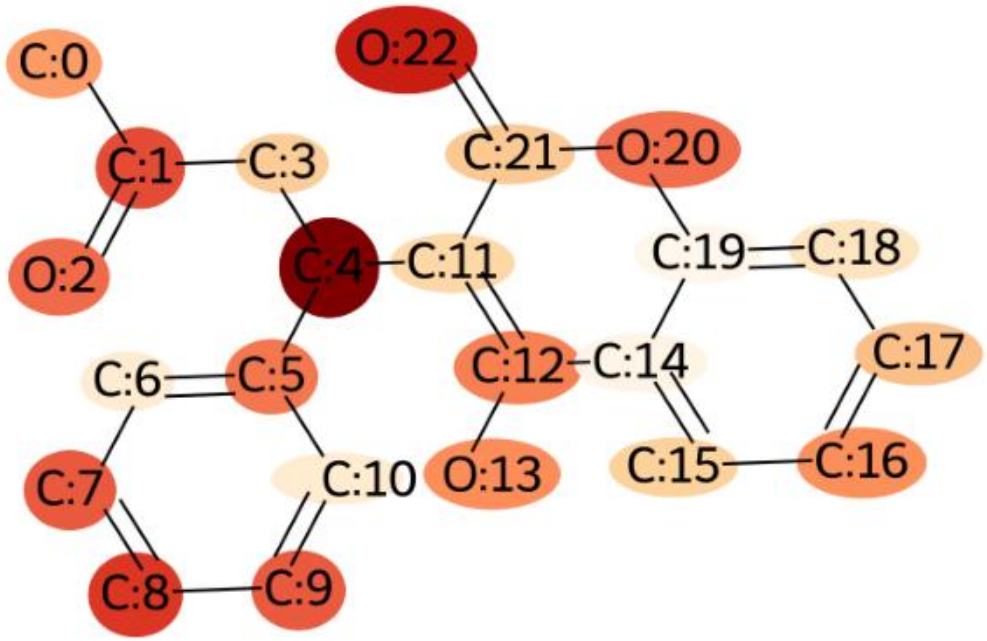
Drug B:
- CC -> a general structural fragment; c -> an aromatic hydrophobic ring

4. Interaction-level interpretation

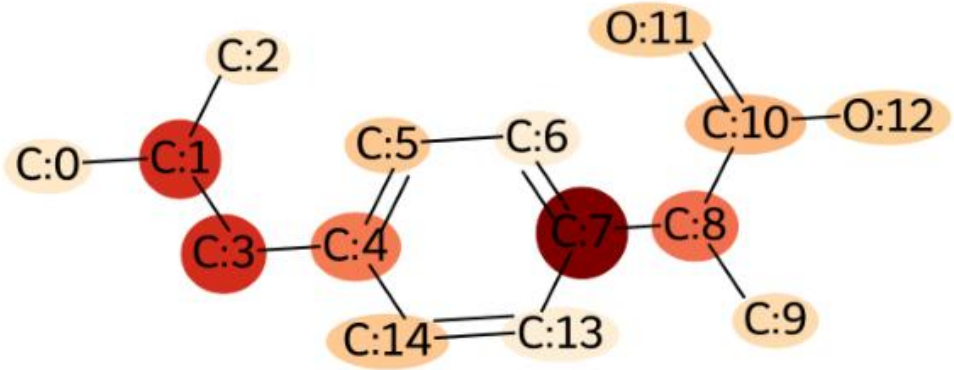
The model highlights an oxygen-rich polar chain or functional group in Warfarin and a general structural fragment in Ibuprofen, suggesting that their interaction may influence toxicity through combined structural and chemical effects.

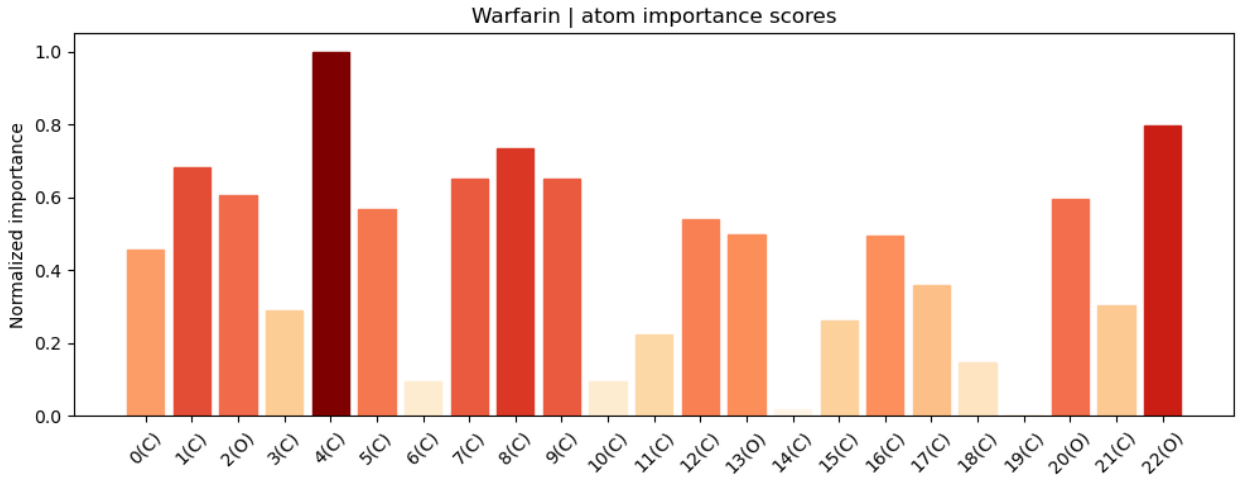
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

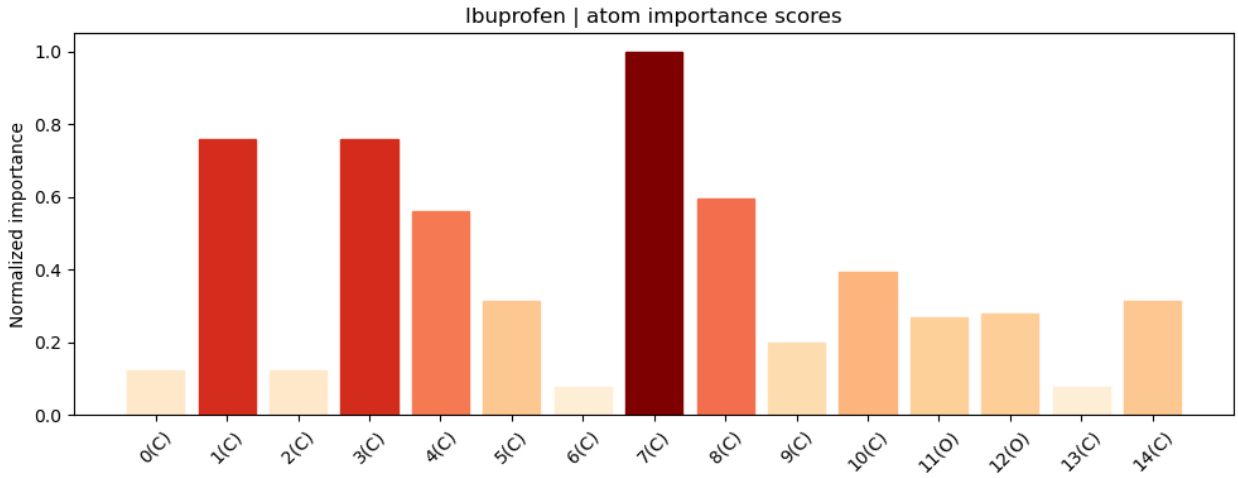
Warfarin | highlighted atom importance



Ibuprofen | highlighted atom importance







Test Case 04 (Vitamin C + Paracetamol)

```
Enter Drug A common name or SMILES: Vitamin C
Enter Drug B common name or SMILES: Paracetamol
Resolved Drug A (name): Vitamin C
SMILES A: O=C1OC(C(O)CO)C(O)=C1O
Resolved Drug B (name): Paracetamol
SMILES B: CC(=O)Nc1ccc(O)cc1

--- Chemical Interpretation ---

Drug A: Vitamin C
Drug B: Paracetamol

Predicted pLD50: 1.7445
Predicted toxicity level: low toxicity

1. Model focus
```

Drug A important atoms:
- C sp3 degree 2 2 H (atom 6); ring C sp2 degree 3 (atom 1); ring C sp3 degree 3 1 H (atom 3)

Drug B important atoms:
- O sp2 degree 1 1 H (atom 8); C sp3 degree 1 3 H (atom 0); C sp2 degree 3 (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 6 is a C atom that may contribute to the local chemical environment. Atom 1 is a ring carbon, which may contribute to structural rigidity. Atom 3 is a ring carbon, which may contribute to structural rigidity.

Drug B:
- Atom 8 is an oxygen atom, which may indicate a polar functional group. Atom 0 is a C atom that may contribute to the local chemical environment. Atom 1 is a highly connected C atom, which may lie in a structurally important region.

3. Key structural fragments

Drug A:
- C -> a non-aromatic ring structure; C -> a general structural fragment

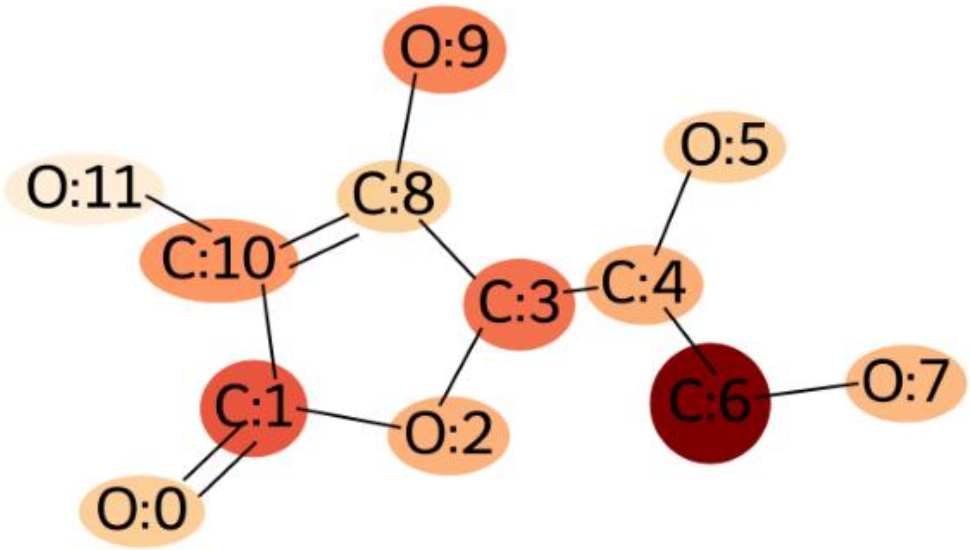
Drug B:
- O -> an oxygen-rich polar chain or functional group

4. Interaction-level interpretation

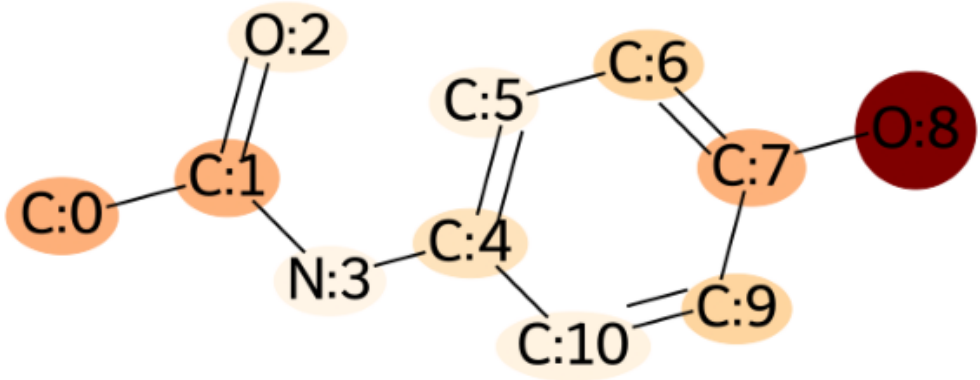
The model highlights an aromatic region in one molecule and a polar or heteroatom-rich region in the other, suggesting that interactions between hydrophobic and polar functionalities may contribute to the observed toxicity.

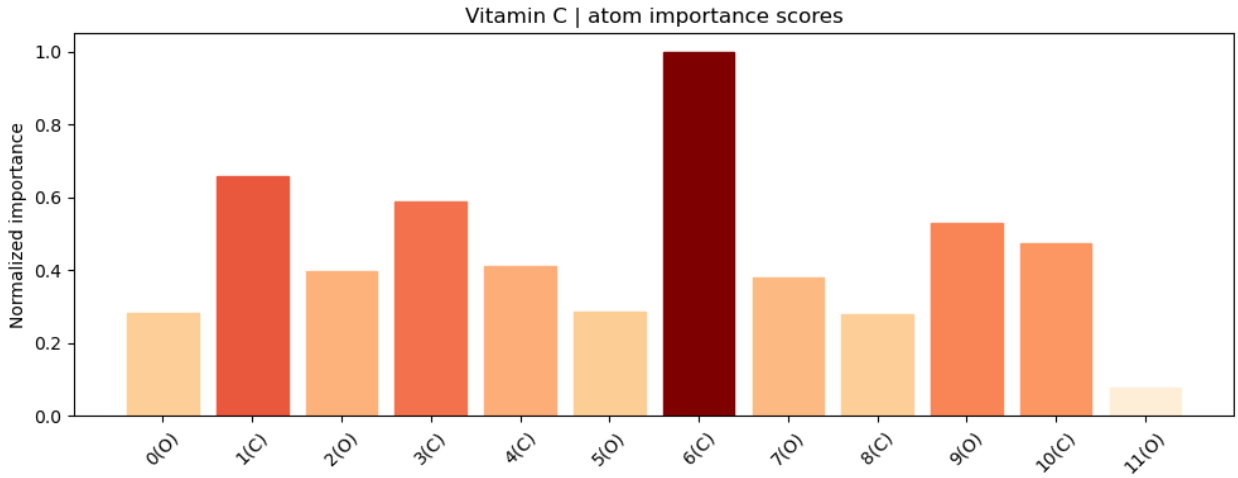
NOTE:
This explanation reflects how the model makes decisions. It does not represent a confirmed biological mechanism.

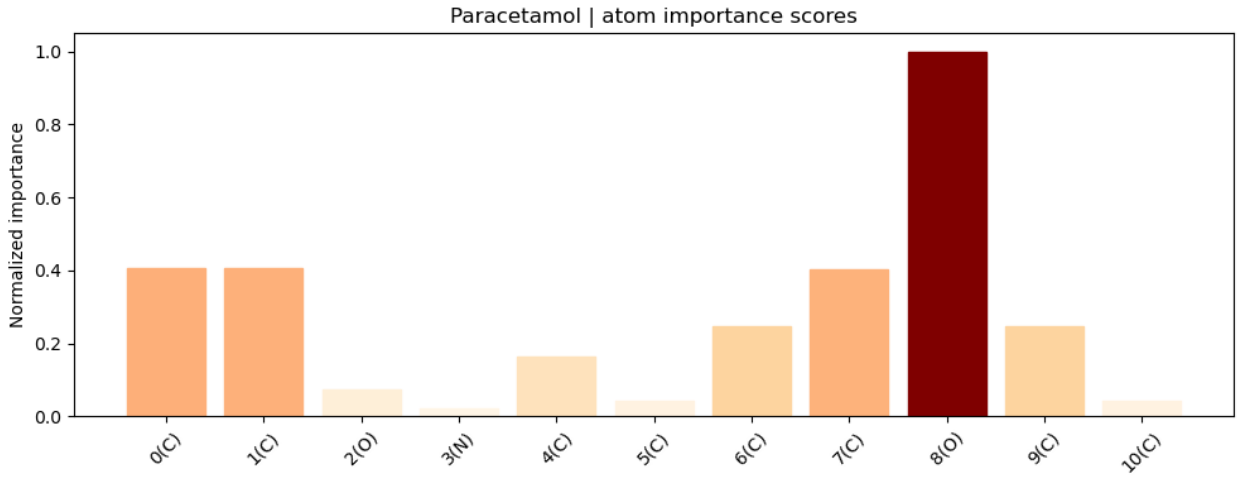
Vitamin C | highlighted atom importance



Paracetamol | highlighted atom importance







Test Case 05 (Tramadol + Sertraline)

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Enter Drug A common name or SMILES: Tramadol
Enter Drug B common name or SMILES: Sertraline
Resolved Drug A (name): Tramadol
SMILES A: COc1cccc(C2(O)CCCCC2CN(C)C)c1
Resolved Drug B (name): Sertraline
SMILES B: CNC1CCC(c2ccc(Cl)c(Cl)c2)c2ccccc21

--- Chemical Interpretation ---

Drug A: Tramadol
Drug B: Sertraline

Predicted pLD50: 2.6709
Predicted toxicity level: high toxicity

1. Model focus
```

Drug A important atoms:
- N sp3 degree 3 (atom 15); O sp2 degree 2 (atom 1); ring C sp3 degree 2 2 H (atom 12)

Drug B important atoms:
- ring C sp3 degree 3 1 H (atom 2); ring C sp3 degree 2 2 H (atom 3); N sp 3 degree 2 1 H (atom 1)

2. Chemical meaning of important atoms

Drug A:
- Atom 15 is a nitrogen atom, which may indicate an amine-like or heteroat om-rich reactive region. Atom 1 is an oxygen atom, which may indicate a po lar functional group. Atom 12 is a ring carbon, which may contribute to st ructural rigidity.

Drug B:
- Atom 2 is a ring carbon, which may contribute to structural rigidity. At om 3 is a ring carbon, which may contribute to structural rigidity. Atom 1 is a nitrogen atom, which may indicate an amine-like or heteroatom-rich re active region.

3. Key structural fragments

Drug A:
- O -> an oxygen-rich polar chain or functional group; N -> an amine-like or nitrogen-containing region

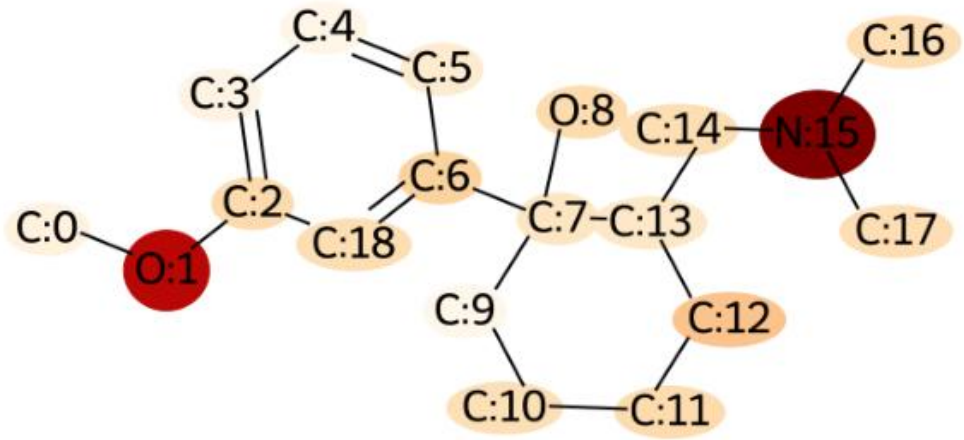
Drug B:
- C -> a non-aromatic ring structure

4. Interaction-level interpretation

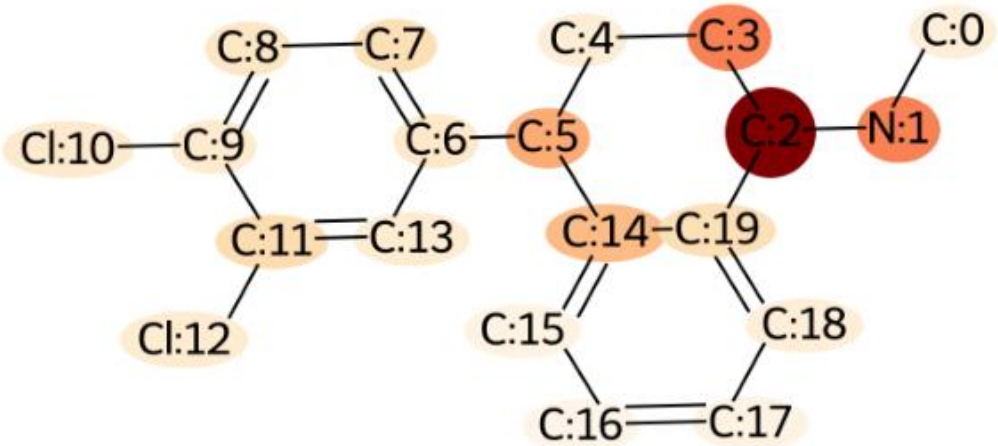
The model highlights an aromatic region in one molecule and a polar or het eroatom-rich region in the other, suggesting that interactions between hyd rophobic and polar functionalities may contribute to the observed toxicity .

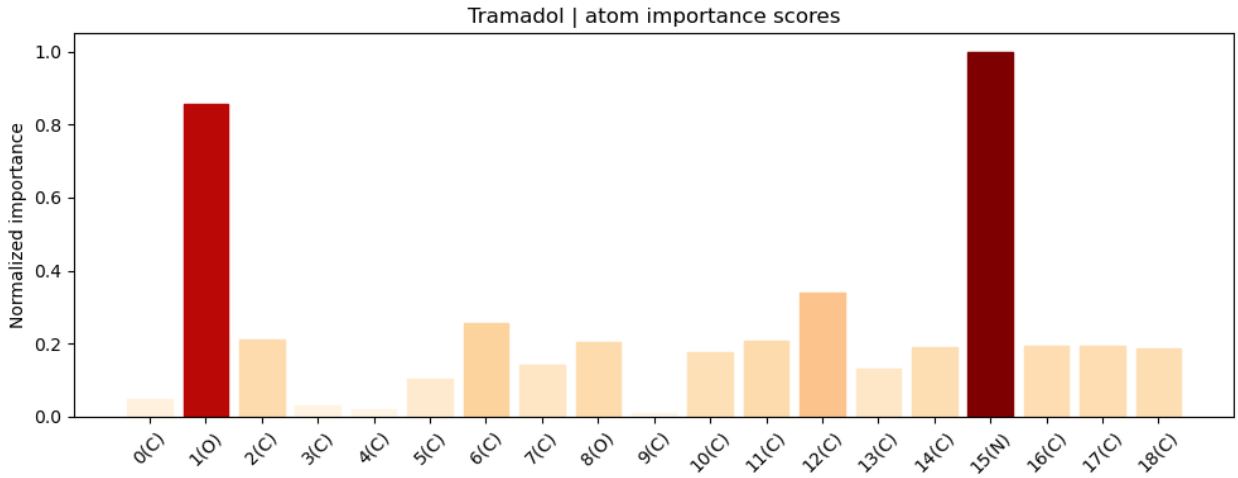
NOTE:
This explanation reflects how the model makes decisions. It does not repre sent a confirmed biological mechanism.

Tramadol | highlighted atom importance



Sertraline | highlighted atom importance




Tramadol | atom importance scores
Normalized importance
1.0
0.8
0.6
0.4
0.2
0.0
0(C)
1(O)
2(C)
3(C)
4(C)
5(C)
6(C)
7(C)
8(O)
9(C)
10(C)
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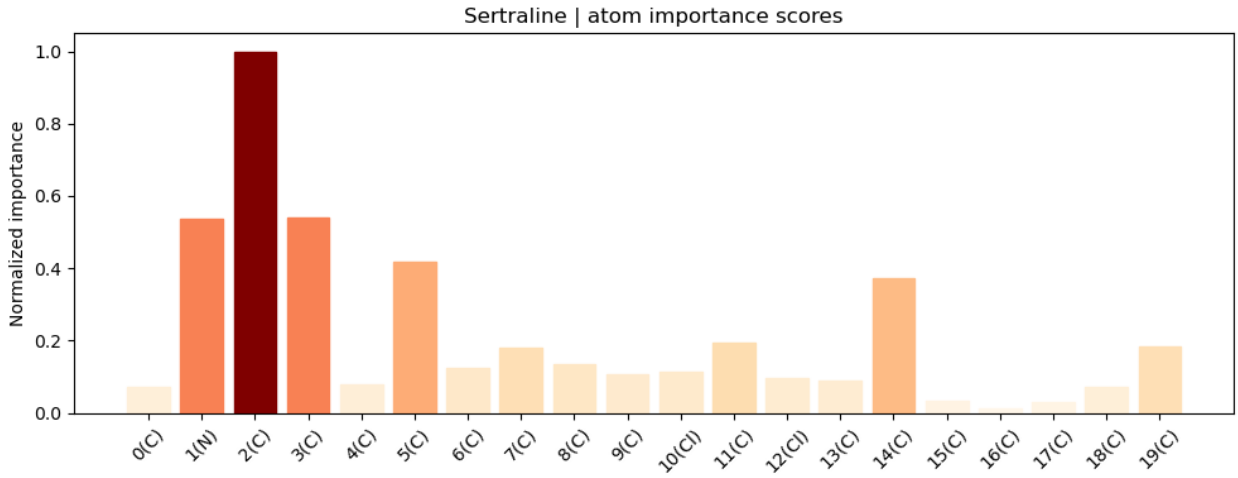
14(C)
15(N)
16(C)
17(C)
18(C)


Sertraline | atom importance scores
Normalized importance
1.0
0.8
0.6
0.4
0.2
0.0
0(C)
1(N)
2(C)
3(C)
4(C)
5(C)
6(C)
7(C)
8(C)
9(C)
10(Cl)
11(C)
12(Cl)
13(C)
14(C)
15(C)
16(C)
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18(C)
19(C)